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VEHICLE LIQUID-COOLING HEAT SINK PLATE HAVING FIN SETS WITH DIFFERENT SURFACE AREAS

Abstract

A vehicle liquid-cooling heat sink plate having fin sets with different surface areas. The vehicle liquid-cooling heat sink plate includes a heat-dissipating plate body and three fin sets. The heat-dissipating plate body has a first heat-dissipating surface and a second heat-dissipating surface that are opposite to each other, the first heat-dissipating surface is used for contacting three traction inverter power component sets, and the second heat-dissipating surface is used for contacting a cooling fluid. The second heat-dissipating surface of the heat-dissipating plate body along a flow direction of the cooling fluid is divided into three heat-dissipating areas which are spaced apart from each other, and the three heat-dissipating areas respectively correspond to three projection areas that are respectively generated by the three traction inverter power component sets.

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Background/Summary

CROSS-REFERENCE TO RELATED PATENT APPLICATION [0001] This application is a Continuation-in-Part of the U.S. patent application Ser. No. 17/963,987, filed on Oct. 11, 2022, and entitled “VEHICLE WATER-COOLING HEAT SINK PLATE HAVING FIN SETS WITH DIFFERENT SURFACE AREAS,” now pending, the entire disclosures of which are incorporated herein by reference.

FIELD OF THE DISCLOSURE

[0002] The present disclosure relates to a vehicle liquid-cooling heat sink plate and a vehicle liquid-cooling radiator, and more particularly to a vehicle liquid-cooling heat sink plate having fin sets with different surface areas, and a closed vehicle liquid-cooling radiator using the same.

BACKGROUND OF THE DISCLOSURE

[0003] Radiators are widely used in various products. Generally speaking, higher-end products usually use water-cooling/liquid-cooling radiators, which have the advantages of quietness and stable cooling comparing with air-cooling radiators, but for automotive power chips that operate faster and faster, the current liquid-cooling radiator is still unable to meet the heat-dissipating requirements of the automotive power chips. Therefore, how to dissipate heat more effectively through the liquid-cooling technology is one of the issues that needs to be solved in the related field.

SUMMARY OF THE DISCLOSURE

[0004] In response to the above-referenced technical inadequacy, the present disclosure provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas, and a closed vehicle liquid-cooling radiator using the same.

[0005] In one aspect, the present disclosure provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas, which includes a heat-dissipating plate body, a first fin set, a second fin set and a third fin set, the heat-dissipating plate body having a first heat-dissipating surface and a second heat-dissipating surface opposite to each other, the first heat-dissipating surface being used for contacting a first, a second and a third traction inverter power component set, the first, the second and the third traction inverter power component set cooperating with each other to form an inverter power module and being used to generate three-phase alternating current for driving a motor of a vehicle, the second heat-dissipating surface being used for contacting a cooling fluid, the second heat-dissipating surface of the heat-dissipating plate body along a flow direction of the cooling fluid being divided into a first, a second and a third heat-dissipating area which are spaced apart from each other and have the same size, the first, the second and the third fin set being respectively located on the first, the second and the third heat-dissipating area, and a first, a second and a third projection area respectively generated by the first, the second and the third traction inverter power component set being projected onto the second heat-dissipating surface respectively defined as the first, the second and the third heat-dissipating area; in which a surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid is less than or equal to a surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid; in which the surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid is less than or equal to a surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid; in which the surface area of the first fin set located in the first heat-dissipating area in contact

with the cooling fluid is less than the surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid; in which any two adjacent ones of the first, second, and third heat-dissipating areas have an auxiliary heat-dissipating area formed therebetween, a guide fin set is formed on the second heat-dissipating surface and located in the auxiliary heat-dissipating area, and a surface area of the guide fin set located in the auxiliary heat-dissipating area in contact with the cooling fluid is less than 50% of the surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid.

[0006] In another aspect, the present disclosure provides a closed vehicle liquid-cooling radiator, which includes a vehicle liquid-cooling heat sink plate having fin sets with different surface areas. The vehicle liquid-cooling heat sink plate includes a heat-dissipating plate body, a first fin set, a second fin set and a third fin set, the heat-dissipating plate body having a first heat-dissipating surface and a second heat-dissipating surface opposite to each other, the first heat-dissipating surface being used for contacting a first, a second and a third traction inverter power component set, the first, the second and the third traction inverter power component set cooperating with each other to form an inverter power module and being used to generate three-phase alternating current for driving a motor of a vehicle, the second heat-dissipating surface being used for contacting a cooling fluid, the second heat-dissipating surface of the heat-dissipating plate body along a flow direction of the cooling fluid being divided into a first, a second and a third heat-dissipating area which are spaced apart from each other and have the same size, the first, the second and the third fin set being respectively located on the first, the second and the third heat-dissipating area, and a first, a second and a third projection area respectively generated by the first, the second and the third traction inverter power component set being projected onto the second heat-dissipating surface respectively defined as the first, the second and the third heat-dissipating area; in which a surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid is less than or equal to a surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid; in which the surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid is less than or equal to a surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid; in which the surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid is less than the surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid. The closed vehicle liquid-cooling radiator further includes a heat-dissipating base, in which the heat-dissipating base has a groove recessed therefrom, and the heat-dissipating base is combined with the heat-dissipating plate body, so that a chamber is formed between the groove of the heat-dissipating base and the second heat-dissipating surface of the heat-dissipating plate body, and the first, the second and the third fin set are located in the chamber.

[0007] In one of the preferred embodiments, a length direction of each of the fins of the guide fin set located in the auxiliary heat-dissipating area is inclined at an included angle of 5 to 25 degrees relative to the flow direction of the cooling fluid.

[0008] In one of the preferred embodiments, a contour of each of the fins of the first fin set that is located in the first heat-dissipating area is drop-shaped, and a contour of each of the fins of the second fin set and the third fin set that are respectively located in the second heat-dissipating area and the third heat-dissipating area is round-shaped.

[0009] In one of the preferred embodiments, a centroid-to-centroid distance between any two adjacent ones of the fins of the first fin set that is located in the first heat-dissipating area ranges from 1.3 mm to 1.5 mm, and a centroid-to-centroid distance between any two adjacent ones of the fins of the third fin set that is located in the third heat-dissipating area ranges from 1.0 mm to 1.2 mm.

[0010] In one of the preferred embodiments, the fins of each of the first, the second and the third fin set have a same height ranging between 2 mm and 10 mm.

[0011] These and other aspects of the present disclosure will become apparent from the following description of the embodiment taken in conjunction with the following drawings and their captions, although variations and modifications therein can be affected without departing from the spirit and scope of the novel concepts of the disclosure.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0012] The described embodiments can be better understood by reference to the following description and the accompanying drawings, in which:

[0013] FIG. 1 is a schematic top view of a first embodiment of the present disclosure;

[0014] FIG. 2 is a schematic bottom view of the first embodiment of the present disclosure;

[0015] FIG. 3 is a schematic bottom view of a second embodiment of the present disclosure;

[0016] FIG. 4 is a schematic bottom view of a third embodiment of the present disclosure;

[0017] FIG. 5 is a schematic bottom view of a fourth embodiment of the present disclosure;

[0018] FIG. 6 is a cross-sectional view of a fin of the present disclosure;

[0019] FIG. 7 is a schematic side view of another fin of the present disclosure;

[0020] FIG. 8 is a cross-sectional view of another fin of the present disclosure;

[0021] FIG. 9 is a cross-sectional view of another fin of the present disclosure;

[0022] FIG. 10 is a schematic side view of a fifth embodiment of the present disclosure;

[0023] FIG. 11 is a schematic bottom view of a sixth embodiment of the present disclosure;

[0024] FIG. 12 is an enlarged view of part XII of FIG. 11; and

[0025] FIG. 13 is an enlarged view of part XIII of FIG. 11.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENTS

[0026] The present disclosure is more particularly described in the following examples that are intended as illustrative only since numerous modifications and variations therein will be apparent to those skilled in the art. Like numbers in the drawings indicate like components throughout the views. As used in the description herein and throughout the claims that follow, unless the context clearly dictates otherwise, the meaning of “a”, “an”, and “the” includes plural reference, and the meaning of “in” includes “in” and “on”. Titles or subtitles can be used herein for the convenience of a reader, which shall have no influence on the scope of the present disclosure.

[0027] The terms used herein generally have their ordinary meanings in the art. In the case of conflict, the present document, including any definitions given herein, will prevail. The same thing can be expressed in more than one way. Alternative language and synonyms can be used for any term(s) discussed herein, and no special significance is to be placed upon whether a term is elaborated or discussed herein. A recital of one or more synonyms does not exclude the use of other synonyms. The use of examples anywhere in this specification including examples of any terms is illustrative only, and in no way limits the scope and meaning of the present disclosure or of any exemplified term. Likewise, the present disclosure is not limited to various embodiments given herein. Numbering terms such as “first”, “second” or “third” can be used to describe various components, signals or the like, which are for distinguishing one component/signal from another one only, and are not intended to, nor should be construed to impose any substantive limitations on the components, signals or the like.

First Embodiment

[0028] Referring to FIG. 1 and FIG. 2, which are one of the embodiments of the present disclosure. The embodiment of the present disclosure provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas, which is used to be arranged in a closed type or a semi-open type vehicle liquid-cooling radiator. As shown in FIG. 1 and FIG. 2, the vehicle liquid-cooling heat sink plate having fin sets with different surface areas (hereinafter referred to as a vehicle liquid-

cooling heat sink plate **M1**) provided according to the embodiment of the present disclosure basically includes a heat-dissipating plate body **10**, and three fin sets **20** (including a first fin set **20a**, a second fin set **20b** and a third fin set **20c**).

[0029] In this embodiment, the heat-dissipating plate body **10** can be made of high thermal conductivity materials, such as copper, copper alloy, aluminum and aluminum alloy. The heat-dissipating plate body **10** has a first heat-dissipating surface **11** and a second heat-dissipating surface **12** opposite to each other. The first heat-dissipating surface **11** is used for contacting a first, a second and a third traction inverter power component set (**30a**, **30b**, **30c**), and the first, the second and the third traction inverter power component set (**30a**, **30b**, **30c**) cooperate with each other to form an inverter power module and are used to generate three-phase alternating current for driving a motor of a vehicle. The second heat-dissipating surface **12** is used for contacting a cooling fluid (e.g., water or ethylene glycol).

[0030] In this embodiment, one end of the heat-dissipating plate body **10** can be located at the upstream side of the cooling fluid, and another end of the heat-dissipating plate body **10** can be located at the downstream side of the cooling fluid, and the second heat-dissipating surface **12** of the heat-dissipating plate body **10** along a flow direction D of the cooling fluid can be divided into three heat-dissipating areas (such as a first, a second and a third heat-dissipating area **13a**, **13b**, **13c**) which are spaced apart from each other and have the same size. Furthermore, the first heat-dissipating area **13a** is located near the upstream side of the cooling fluid, the third heat-dissipating area **13c** is located near the downstream side of the cooling fluid, and the second heat-dissipating area **13b** is located between the first heat-dissipating area **13a** and the third heat-dissipating area **13c**.

[0031] In this embodiment, the first, the second and the third fin set (**20a**, **20b**, **20c**) are respectively located on the first, the second and the third heat-dissipating area (**13a**, **13b**, **13c**) of the second heat-dissipating surface **12**. The first, the second and the third fin set (**20a**, **20b**, **20c**) are preferably integrally connected to the second heat-dissipating surface **12** of the heat-dissipating plate body **10** by metal injection molding (MIM), but can also be formed on the second heat-dissipating surface **12** of the heat-dissipating plate body **10** by forging, or can also be connected to the second heat-dissipating surface **12** of the heat-dissipating plate body **10** by welding or assembling. In addition, a first, a second and a third projection area (**301**, **302**, **303**) respectively generated by the first, the second and the third traction inverter power component set (**30a**, **30b**, **30c**) are projected onto the second heat-dissipating surface **12** respectively defined as the first, the second and the third heat-dissipating area (**13a**, **13b**, **13c**). Furthermore, the surface area of the first fin set **20a** located in the first heat-dissipating area **13a** in contact with the cooling fluid is less than or equal to ($\leq$) the surface area of the second fin set **20b** located in the second heat-dissipating area **13b** in contact with the cooling fluid, the surface area of the second fin set **20b** located in the second heat-dissipating area **13b** in contact with the cooling fluid is less than or equal to ($\leq$) the surface area of the third fin set **20c** located in the third heat-dissipating area **13c** in contact with the cooling fluid, and the surface area of the first fin set **20a** located in the first heat-dissipating area **13a** in contact with the cooling fluid is less than the surface area of the third fin set **20c** located in the third heat-dissipating area **13c** in contact with the cooling fluid. Therefore, the fluid velocity in the area close to the upstream side can be slowed down to reduce friction, so that the overall pressure drop will not drop too much, which can avoid increasing the operating energy consumption of the liquid pump, and can ensure the temperature uniformity of the three traction inverter power component sets during the heat dissipating and cooling process.

[0032] In this embodiment, the fins **21** of each fin set **20** are each a plate-shaped fin and arranged in parallel, and the fins **21** of each fin set **20** have the same size, but the number of the fins **21** of the first fin set **20a** is fewer than the number of fins **21** of the second fin set **20b**, so that the surface area of the first fin set **20a** in contact with the cooling fluid is less than the surface area of the second fin set **20b** in contact with the cooling fluid. In addition, the number of fins **21** of the second

fin set **20b** is fewer than the number of fins **21** of the third fin set **20c**, so that the surface area of the second fin set **20b** in contact with the cooling fluid is less than the surface area of the third fin set **20c** in contact with the cooling fluid. Furthermore, the length direction of each fin **21** of each fin set **20** is parallel to the flow direction **D** of the cooling fluid.

Second Embodiment

[0033] Referring to FIG. 3, which is a second embodiment of the present disclosure. This embodiment is substantially the same as the first embodiment, and the differences are described as follows.

[0034] This embodiment provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas (hereinafter referred to as a vehicle liquid-cooling heat sink plate **M2**). In this embodiment, the length direction of each fin **21** of the first fin set **20a** located in the first heat-dissipating area **13a** is parallel to the flow direction **D** of the cooling fluid or inclined at an included angle fewer than 5 degrees relative to the flow direction **D** of the cooling fluid. The length direction of each fin **21** of the second fin set **20b** located in the second heat-dissipating area **13b** is inclined at an included angle of 5 to 20 degrees relative to the flow direction **D** of the cooling fluid. The length direction of each fin **21** of the third fin set **20c** located in the third heat-dissipating area **13c** is inclined at an included angle of 5 to 25 degrees relative to the flow direction **D** of the cooling fluid, and the inclination direction of each fin **21** of the second fin set **20b** are different from the inclination direction of each fin **21** of the third fin set **20c**, thereby guiding the cooling fluid and increasing the contact time of the cooling fluid with the fins **21**.

Third Embodiment

[0035] Referring to FIG. 4, which is a third embodiment of the present disclosure. This embodiment is substantially the same as the first embodiment, and the differences are described as follows.

[0036] This embodiment provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas (hereinafter referred to as a vehicle liquid-cooling heat sink plate **M3**). In this embodiment, the fins **21** of each fin set **20** have the same size, but the number of fins **21** of the first fin set **20a** is fewer than the number of the fins **21** of the second fin set **20b**, so that the surface area of the first fin set **20a** in contact with the cooling fluid is less than the surface area of the second fin set **20b** in contact with the cooling fluid. In addition, the number of the fins **21** of the second fin set **20b** is equal to the number of the fins **21** of the third fin set **20c**, so that the surface area of the second fin set **20b** in contact with the cooling fluid is equal to the surface area of the third fin set **20c** in contact with the cooling fluid.

Fourth Embodiment

[0037] Referring to FIG. 5, which is the fourth embodiment of the present disclosure. This embodiment is substantially the same as the first embodiment, and the differences are described as follows.

[0038] The present embodiment provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas (hereinafter referred to as a vehicle liquid-cooling heat sink plate **M4**). In this embodiment, the three fin sets **20** are respectively located in the three heat-dissipating areas, and the three projection areas respectively generated by the three spaced traction inverter power component sets are projected onto the second heat-dissipating surface **12** respectively defined as three heat-dissipating areas (such as a first, a second and a third heat-dissipating area (**13a**, **13b**, **13c**)). Furthermore, the first fin set **20a**, the second fin set **20b**, and the third fin set **20c** of the three fin sets **20** are located in the first heat-dissipating area **13a**, the second heat-dissipating area **13b**, and the third heat-dissipating area **13c**, respectively. Moreover, the surface area of the first fin set **20a** located in the first heat-dissipating area **13a** in contact with the cooling fluid is less than or equal to ($\leq$) the surface area of the second fin set **20b** located in the second heat-dissipating area **13b** in contact with the cooling fluid, the surface area of the second fin set **20b** located in the second heat-dissipating area **13b** in contact with the cooling fluid is less than or equal to ($\leq$) the

surface area of the third fin set **20c** located in the third heat-dissipating area **13c** in contact with the cooling fluid, and the surface area of the first fin set **20a** located in the first heat-dissipating area **13a** in contact with the cooling fluid is less than the surface area of the third fin set **20c** located in the third heat-dissipating area **13c** in contact with the cooling fluid.

[0039] In this embodiment, the fins **21** of each fin set **20** are each a pin-shaped fin and arranged in a high density (that is to say, the fins **21** of each fin set **20** are adjacent to each other), but the fin sets **20** have different arrangement density. The arrangement density of the first fin set **20a** is smaller than the arrangement density of the second fin set **20b**, so that the surface area of the first fin set **20a** in contact with the cooling fluid is less than the surface area of the second fin set **20b** in contact with the cooling fluid. In addition, the arrangement density of the second fin set **20a** is smaller than the arrangement density of the third fin set **20c** (that is to say, the arrangement density of the third fin set **20c** is the highest), so that the surface area of the second fin set **20b** in contact with the cooling fluid is less than the surface area of the third fin set **20c** in contact with the cooling fluid. More particularly, the surface area of the first fin set **20a** in contact with the cooling fluid is between 10000 mm² and 11500 mm², the surface area of the second fin set **20b** in contact with the cooling fluid is between 11500 mm² and 13000 mm², and the surface area of the third fin set **20c** in contact with the cooling fluid is between 13000 mm² and 14500 mm².

[0040] Furthermore, the cross-section of the pin-shaped fin in each heat-dissipating area can be circular, square, diamond, drop-shaped, oval or other shape. In this embodiment, the cross-section of each fin **21** located in the first heat-dissipating area **13a** is preferably an ellipse, the cross-section of each fin **21** located in the second heat-dissipating area **13b** is circular, and the cross-section of each fin **21** located in the third heat-dissipating area **13c** is also circular. The fins **21** of each fin set **20** have the same height preferably ranging between 2 mm and 10 mm. In one embodiment, as shown in FIG. 6, the cross-section of the fin **21** can be a non-circular shape that has a round radius R ranging between 0.1 mm and 1 mm. In addition, as shown in FIG. 7, the draft angle θ of the fin **21** needs to be fewer than 5 degrees. In another embodiment, as shown in FIG. 8 and FIG. 9, the cross-section of the fin **21** can be drop-shaped or wing-shaped.

Fifth Embodiment

[0041] Referring to FIG. 10, which is a fifth embodiment of the present disclosure. This embodiment is substantially the same as the first embodiment, and the differences are described as follows.

[0042] In this embodiment, a closed vehicle liquid-cooling radiator **M5** is provided. Furthermore, the closed vehicle liquid-cooling radiator **M5** of this embodiment includes a heat-dissipating base **40** and a vehicle liquid-cooling heat sink plate having fin sets with different surface areas as described in any of the foregoing embodiments. The heat-dissipating base **40** of the present embodiment has a groove **41** recessed therefrom, and the heat-dissipating base **40** is combined with the heat-dissipating plate body **10**, so that a chamber **CH** is formed between the groove **41** of the heat-dissipating base **40** and the second heat-dissipating surface **12** of the heat-dissipating plate body **10**, and the first fin set **20a**, the second fin set **20b**, and the third fin set **20c** are located in the chamber **CH**. Furthermore, the heat-dissipating base **40** is further formed with a water inlet through hole **401** and a water outlet through hole **402** respectively communicated with the chamber **CH**, so that the cooling fluid can flow into the chamber **CH** through the water inlet through hole **401** and flow out of the chamber **CH** through the water outlet through hole **402**, in order to further improve the heat-dissipating performance through the closed fluid circulation loop.

Sixth Embodiment

[0043] Referring to FIG. 11, FIG. 12, and FIG. 13, which are one of the embodiments of the present disclosure. This embodiment is substantially the same as the first embodiment, and the differences are described as follows.

[0044] The present embodiment provides a vehicle liquid-cooling heat sink plate having fin sets with different surface areas (hereinafter referred to as a vehicle liquid-cooling heat sink plate **M6**).

In this embodiment, any two adjacent ones of the first, second, and third heat dissipation areas **13a**, **13b**, and **13c** have an auxiliary heat-dissipating area **14** formed therebetween, and a guide fin set **40** is formed on the second heat-dissipating surface **12** and located in the auxiliary heat-dissipating area **14**. Furthermore, a surface area of the guide fin set **40** located in the auxiliary heat-dissipating area **14** in contact with the cooling fluid is less than 50% of the surface area of the first fin set **20a** located in the first heat-dissipating area **13a** in contact with the cooling fluid. Accordingly, pressure drop can be minimized at both an area that is near an upstream side of a liquid flow and an auxiliary heat-dissipating area, such that an excessive pressure drop does not occur, and an operating energy consumption of a liquid pump can be prevented from being increased.

Furthermore, by the design of the auxiliary heat-dissipating area, the excessive pressure drop does not occur and fluids having different temperatures can be mixed together, such that temperature homogeneity of the first, second and third traction inverter power component sets **30a**, **30b**, and **30c** that are spaced apart from each other can be maintained in a process of heat dissipation.

[0045] Moreover, a length direction of each fin **41** of the guide fin set **40** located in the auxiliary heat-dissipating area **14** is inclined at an included angle of 5 to 25 degrees relative to the flow direction D of the cooling fluid, such that a flow of the cooling fluid is improved by being guided by the guide fin set **40**.

[0046] In this embodiment, a contour of each fin **21** of the first fin set **20a** that is located in the first heat-dissipating area **13a** is drop-shaped, a contour of each fin **21** of the second fin set **20b** that is located in the second heat-dissipating area **13b** is round-shaped, and a contour of each fin **21** of the third fin set **20c** that is located in the third heat-dissipating area **13c** is round-shaped. Accordingly, the heat dissipation efficiency of the first heat-dissipating area **13a** might be slightly reduced by the drop-shaped fins being arranged in the first heat-dissipating area **13a**; however, the cooling fluid can be smoothly guided to the second and third heat dissipation areas **13b** and **13c** and the overall pressure drop can be reduced by the drop-shaped fins being arranged in the first heat-dissipating area **13a**. Further, through the round-shaped fins with better heat dissipation efficiency being arranged in the second and third heat dissipation areas **13b** and **13c**, the temperature difference between the first heat-dissipating area **13a** (the relative low temperature area) and the third heat-dissipating area **13c** (the relative high temperature area) can be reduced. That is to say, by the drop-shaped fins being arranged in the first heat-dissipating area **13a** and the round-shaped fins being arranged in the second and third heat dissipation areas **13b** and **13c**, the overall pressure drop and the temperature difference can be both reduced at the same time.

[0047] Furthermore, in order to reduce the overall pressure drop and the temperature difference more effectively, the centroid-to-centroid distance B1 between any two adjacent ones of the fins **21** of the first fin set **20a** that is located in the first heat-dissipating area **13a** ranges from 1.3 mm to 1.5 mm, and the centroid-to-centroid distance B2 between any two adjacent ones of the fins **21** of the third fin set **20c** that is located in the third heat-dissipating area **13c** ranges from 1.0 mm to 1.2 mm and is shorter than the centroid-to-centroid distance B1.

[0048] In conclusion, in the vehicle liquid-cooling heat sink plate having fin sets with different surface areas, and the closed vehicle liquid-cooling radiator using the same provided by the present disclosure, by virtue of “the heat-dissipating plate body,” “the first, the second and the third fin set,” “the heat-dissipating plate body having a first heat-dissipating surface and a second heat-dissipating surface opposite to each other, the first heat-dissipating surface being used for contacting a first, a second and a third traction inverter power component set, the first, the second and the third traction inverter power component set cooperating with each other to form an inverter power module and being used to generate three-phase alternating current for driving a motor of a vehicle, and the second heat-dissipating surface being used for contacting a cooling fluid,” “the second heat-dissipating surface of the heat-dissipating plate body along a flow direction of the cooling fluid being divided into a first, a second and a third heat-dissipating area which are spaced apart from each other and have the same size,” “the first, the second and the third fin set being

respectively located on the first, the second and the third heat-dissipating area, and a first, a second and a third projection area respectively generated by the first, the second and the third traction inverter power component set being projected onto the second heat-dissipating surface respectively defined as the first, the second and the third heat-dissipating area,” “a surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid being less than or equal to a surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid,” “the surface area of the second fin set located in the second heat-dissipating area in contact with the cooling fluid being less than or equal to a surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid,” and “the surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid being less than the surface area of the third fin set located in the third heat-dissipating area in contact with the cooling fluid,” the overall fluid pressure drop will not drop too much, which can avoid increasing the operating energy consumption of the liquid pump, and can ensure the temperature uniformity of the three traction inverter power component sets during the heat dissipating and cooling process.

[0049] The foregoing description of the exemplary embodiments of the disclosure has been presented only for the purposes of illustration and description and is not intended to be exhaustive or to limit the disclosure to the precise forms disclosed. Many modifications and variations are possible in light of the above teaching.

[0050] The embodiments were chosen and described in order to explain the principles of the disclosure and their practical application so as to enable others skilled in the art to utilize the disclosure and various embodiments and with various modifications as are suited to the particular use contemplated. Alternative embodiments will become apparent to those skilled in the art to which the present disclosure pertains without departing from its spirit and scope.

Claims

1. A vehicle liquid-cooling heat sink plate having fin sets with different surface areas, comprising: a heat-dissipating plate body having a first heat-dissipating surface and a second heat-dissipating surface opposite to each other, wherein the first heat-dissipating surface is in contact with a first, a second and a third traction inverter power component set, and the first, the second and the third traction inverter power component set cooperate with each other to form an inverter power module and are configured to generate three-phase alternating current for driving a motor of a vehicle, wherein the second heat dissipation surface is in contact with a cooling liquid, and the second heat-dissipating surface of the heat-dissipating plate body along a flow direction of the cooling liquid is divided into a first, a second and a third heat-dissipating area which are spaced apart from each other and have the same size; a first fin set located on the first heat-dissipating area, wherein the first heat-dissipating area is defined by a first projection area formed by projecting the first traction inverter power component set on the second heat dissipation surface; a second fin set located on the second heat-dissipating area, wherein the second heat-dissipating area is defined by a second projection area formed by projecting the second traction inverter power component set on the second heat dissipation surface; and a third fin set located on the third heat-dissipating area, wherein the third heat-dissipating area is defined by a third projection area formed by projecting the third traction inverter power component set on the second heat dissipation surface; wherein a surface area of the first fin set located in the first heat-dissipating area in contact with the cooling liquid is less than or equal to a surface area of the second fin set located in the second heat-dissipating area in contact with the cooling liquid; wherein the surface area of the second fin set located in the second heat-dissipating area in contact with the cooling liquid is less than or equal to a surface area of the third fin set located in the third heat-dissipating area in contact with the cooling liquid; wherein the surface area of the first fin set located in the first heat-dissipating area in contact with the cooling liquid is less than the surface area of the third fin set located in the third

heat-dissipating area in contact with the cooling liquid; wherein any two adjacent ones of the first, second, and third heat-dissipating areas have an auxiliary heat-dissipating area formed therebetween, a guide fin set is formed on the second heat-dissipating surface and located in the auxiliary heat-dissipating area, and a surface area of the guide fin set located in the auxiliary heat-dissipating area in contact with the cooling fluid is less than 50% of the surface area of the first fin set located in the first heat-dissipating area in contact with the cooling fluid.

2. The vehicle liquid-cooling heat sink plate according to claim 1, wherein a length direction of each of fins of the guide fin set located in the auxiliary heat-dissipating area is inclined at an included angle of 5 to 25 degrees relative to the flow direction of the cooling fluid.

3. The vehicle liquid-cooling heat sink plate according to claim 1, wherein a contour of each of fins of the first fin set that is located in the first heat-dissipating area is drop-shaped, and a contour of each of fins of the second fin set and the third fin set that are respectively located in the second heat-dissipating area and the third heat-dissipating area is round-shaped.

4. The vehicle liquid-cooling heat sink plate according to claim 3, wherein a centroid-to-centroid distance between any two adjacent ones of the fins of the first fin set that is located in the first heat-dissipating area ranges from 1.3 mm to 1.5 mm, and a centroid-to-centroid distance between any two adjacent ones of the fins of the third fin set that is located in the third heat-dissipating area ranges from 1.0 mm to 1.2 mm.

5. The vehicle liquid-cooling heat sink plate according to claim 1, wherein fins of each of the first, the second and the third fin set have a same height ranging between 2 mm and 10 mm.
